ARCM — Autonomous Risk Classification Module

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Modul5 ARCM — Autonomous Risk Classification Module

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Governed Space: Autonomous Risk Classification

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(OEAS) · Operational Authority Integrity Standard (OAIS) · Operational Constraint Integrity Standard (OCNS) · Runtime Integrity Standard (RIS) · INTEGROS® — Integrity Standard · Autonomous Systems · Multi-Agent Infrastructures

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Canonical Language: English (UCL)

Canonical Definition

Autonomous Risk Classification Module (ARCM) defines the structural conditions under which autonomous operational activities, execution environments, agents, systems, authorities, tools, decisions, and consequence-bearing operational states remain materially classifiable according to governance-valid risk conditions across autonomous operational environments. ARCM governs autonomous risk classification. The module ensures that operational risk becomes visible, structured, comparable, governable, and actionable before execution occurs.

A system satisfies ARCM only if:

- risk remains classifiable
- classification criteria remain traceable

- classification logic remains governable
- risk categories remain operationally meaningful
- risk-state transitions remain detectable
- consequence-bearing risk remains visible
- A system that executes actions without governance-valid risk classification does not satisfy ARCM.

Module Operational Role

ARCM defines the risk-classification layer of ORTIS. Its role is to preserve governance-valid visibility of operational risk before, during, and after execution. ARCM answers a foundational governance question:

What level of risk does this operational action actually represent?

Module Operational Space

- ARCM governs:
- risk classification
- risk categorization
- risk visibility
- operational risk tiers
- execution-risk assessment
- authority-risk alignment
- consequence-bearing risk mapping
- runtime risk-state classification
- The module applies wherever autonomous systems perform actions capable of creating operational consequences.

Module Function

- The module applies wherever systems must preserve:
- risk visibility
- risk categorization

governance-valid execution assessment consequence-bearing risk awareness operational risk traceability trust-risk alignment Its function is to ensure that operational risk becomes classifiable before governance decisions are made.

Minimum Implementation Framework

1. Define the Risk Classification Object

The organization must define which operational objects require risk classification. This may include: autonomous agents execution tasks tool usage authority actions operational workflows decision systems consequence-bearing activities multi-agent coordination actions

2. Define Risk Classification Conditions

The system must define the conditions under which risk categories are assigned. This includes: risk severity criteria consequence thresholds sensitivity conditions authority requirements operational impact conditions trust-related risk conditions

3. Define Risk Classification Detection Logic

The system must define how risk classifications are determined and updated. This may include: consequence analysis operational impact analysis sensitivity assessment authority analysis trust-state assessment runtime context evaluation escalation-condition detection

4. Define Operational Response or Governance Logic

The system must define governance logic based on risk classifications. Governance response may include: permission assignment execution restriction authority requirements escalation activation monitoring enhancement

trust adjustments operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

- The system must preserve reconstructable traceability of:
- risk classifications
- classification changes
- governance decisions
- execution permissions
- escalation events
- consequence-bearing outcomes
- A system must not remain risk-valid if operational activities cannot be classified according to governance-valid risk
- conditions.

Use Case 1 — Autonomous Enterprise Agent

Environment

Scenario

A large enterprise operates hundreds of autonomous agents performing workflow automation, tool access, decision support, and operational execution.

Application

ARCM classifies operational activities according to consequence-bearing risk levels before execution eligibility is granted.

Result

The organization gains stronger governance visibility, more accurate permission assignment, and improved operational risk control.

Use Case 2 — Autonomous Industrial Operations

Infrastructure

Scenario

A distributed industrial environment coordinates autonomous systems responsible for production control, logistics, resource allocation, and operational optimization.

Application

ARCM classifies operational actions according to potential impact, sensitivity, authority requirements, and consequencebearing risk exposure.

Result

The environment gains stronger risk awareness, improved governance decision-making, and reduced exposure to uncontrolled operational consequences.

Canonical Closing Statement

Autonomous Risk Classification Module (ARCM) defines the structural conditions under which autonomous operational activities, execution environments, agents, systems, authorities, tools, decisions, and consequence-bearing operational states remain materially classifiable according to governance-valid risk conditions across autonomous operational environments. Operational risk cannot be governed if it cannot first be classified. Autonomous execution becomes governancevalid only when risk remains visible, structured, traceable, classifiable, and operationally actionable throughout runtime operation.

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Canonical Source

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